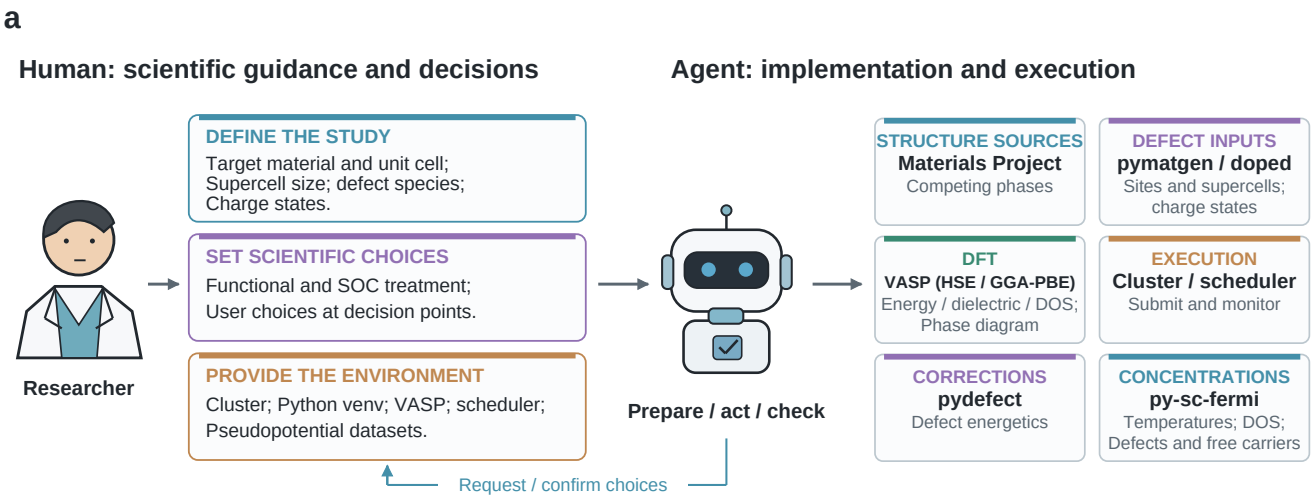
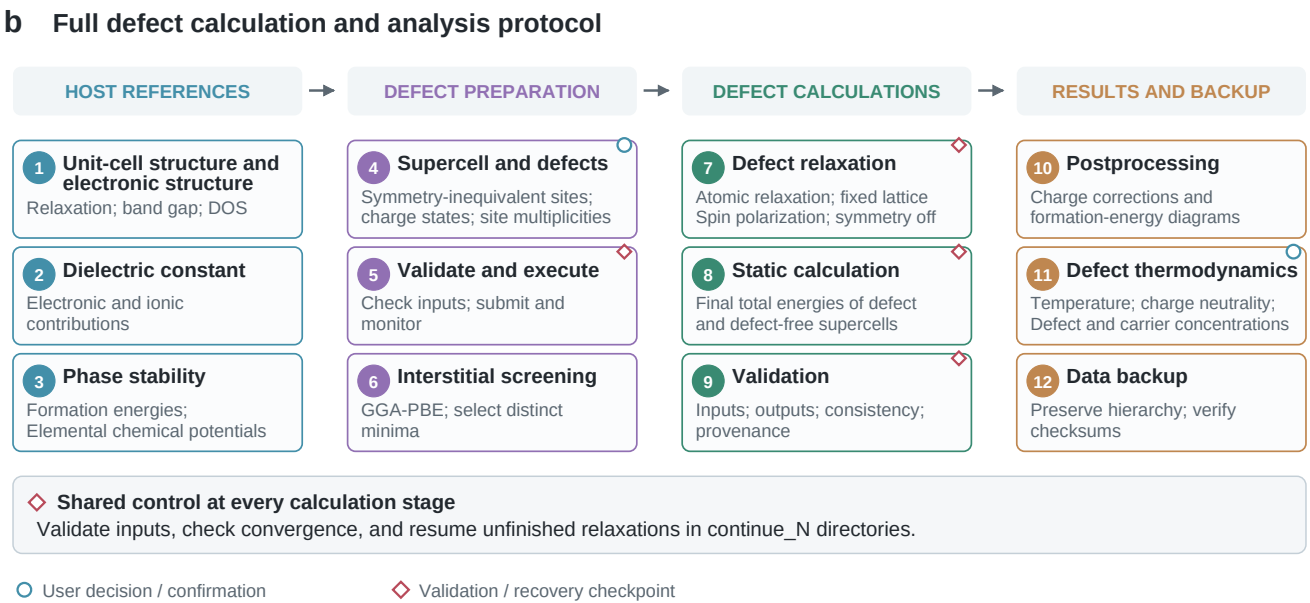

a
Human: scientific guidance and decisions
Agent: implementation and execution
Researcher
DEFINE THE STUDY
Target material and unit cell;
Supercell size; defect species;
Charge states.
SET SCIENTIFIC CHOICES
Functional and SOC treatment;
User choices at decision points.
PROVIDE THE ENVIRONMENT
Cluster; Python venv; VASP; scheduler;
Pseudopotential datasets.
Prepare / act / check
Request / confirm choices
STRUCTURE SOURCES
Materials Project
Competing phases
DEFECT INPUTS
pymatgen / doped
Sites and supercells;
charge states
DFT
VASP (HSE / GGA-PBE)
Energy / dielectric / DOS;
Phase diagram
EXECUTION
Cluster / scheduler
Submit and monitor
CORRECTIONS
pydefect
Defect energetics
CONCENTRATIONS
py-sc-fermi
Temperatures; DOS;
Defects and free carriers


b
Full defect calculation and analysis protocol
HOST REFERENCES
DEFECT PREPARATION
DEFECT CALCULATIONS
RESULTS AND BACKUP
1 Unit-cell structure and electronic structure
Relaxation; band gap; DOS
2 Dielectric constant
Electronic and ionic contributions
3 Phase stability
Formation energies;
Elemental chemical potentials
4 Supercell and defects
Symmetry-inequivalent sites;
charge states; site multiplicities
5 Validate and execute
Check inputs; submit and monitor
6 Interstitial screening
GGA-PBE; select distinct minima
7 Defect relaxation
Atomic relaxation; fixed lattice
Spin polarization; symmetry off
8 Static calculation
Final total energies of defect and defect-free supercells
9 Validation
Inputs; outputs; consistency; provenance
10 Postprocessing
Charge corrections and formation-energy diagrams
11 Defect thermodynamics
Temperature; charge neutrality;
Defect and carrier concentrations
12 Data backup
Preserve hierarchy; verify checksums
Shared control at every calculation stage
Validate inputs, check convergence, and resume unfinished relaxations in continue_N directories.
User decision / confirmation
Validation / recovery checkpoint